

Hans William Alexander Hanley 汉斯·汉隶

Natural Language Processing • AI Safety • Multilingual Representation Learning

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I build scalable methods to interpret and evaluate large language models, including multilingual representation learning, semantic modeling, and LLM agents for high-fidelity content analysis. My PhD research uncovered latent structure in massive text ecosystems, aligned model behavior across languages and domains, and advanced safer, more reliable AI systems.

EDUCATION

Stanford University **PhD, Computer Science, 2020-2025 | MS, Computer Science, 2020-2024**
GPA: 4.02; Advisor: Zakir Durumeric; Meta Research PhD Fellow; NSF GRFP; Stanford EDGE Doctoral Fellow
University of Oxford **MSc, Statistical Science, 2019-2020 | MSc, Computer Science, 2018-2019**
Distinction (Highest Honors); Daniel M. Sachs Class of 1960 Graduating Scholar at Worcester College
Princeton University **BSE, Electrical Engineering, 2014-2018**
Highest Honors, GPA: 3.982; Information Security & Privacy Concentration; Phi Beta Kappa; Tau Beta Pi; Sigma Xi

PROFESSIONAL EXPERIENCE

Microsoft AI, Member of Technical Staff **April 2026–Present**
• Researched LLM-as-a-judge evaluators for online assessment of Human-AI interactions in Microsoft Copilot, measuring task completion, intent satisfaction, reasoning complexity, response quality, and failure modes to guide product hillclimbing.
• Built LLM-based classifiers and measurement frameworks linking customer feedback and user behavior to offline AI evaluations, identifying representativeness and failure-coverage gaps; eval creation and AI evaluation strategy presented to Microsoft AI's CEO.
Meta, Research Scientist, CAS / Central Applied Sciences **March 2025–April 2026**
• Built and deployed large-scale multilingual and multimodal LLM pipelines modeling content semantics, creator verticals, and interaction behavior across Instagram for downstream recommender systems.
• Developed LLM-based evaluations and production metrics for political content quality, reply/comment impact, safety and integrity risks, and downstream user experience.
Google Research, Health AI Student Researcher **Summer 2024**
• Designed LLM agents for structured querying and high-fidelity reasoning over large clinical datasets, with verification layers for failures under distribution shifts.
• Built evaluations of accuracy, safety, and adversarial robustness for LLM-mediated medical data interactions.

RESEARCH HIGHLIGHTS & SELECTED PUBLICATIONS

Tracking the Takes and Trajectories of News Narratives **USENIX Security 2025**
• Built a large-scale narrative-tracking system combining specially fine-tuned (e.g., LoRA) encoder-LLM semantic similarity models and zero-shot stance detection to track 146K narratives across 4K+ news sites and understand narrative pathways (e.g., anti-vaccine and anti-Ukraine influence campaigns).
Across the Firewall: Foreign Media's Influence on Chinese Social Media **PNAS 2025**
• Quantified foreign-media penetration into Chinese platforms using cross-lingual alignment pipelines to detect narrative borrowing on the Russo-Ukrainian War.
Hierarchical Level-Wise News Article Clustering via Multilingual Matryoshka Embeddings **ACL 2025**
• Proposed multilingual Matryoshka embeddings enabling controllable semantic granularity (coarse-to-fine); achieved SOTA on SemEval 2022 Task 8 and built scalable hierarchical clustering pipelines surfacing stories, narratives, and macro-themes.
TATA: Stance Detection via Topic-Agnostic and Topic-Aware Embeddings **EMNLP 2023**
• Developed contrastive embedding models for topic-agnostic and topic-aware stance detection; achieved SOTA zero-shot performance (0.771 F1 on VAST) and improved cross-topic generalization in multi-domain news corpora.

SKILLS

Programming Python, Java, C, C++, C#, Go, SQL, R, MATLAB; Git; PyTorch, TensorFlow, Hugging Face
ML/LLM Transformers; LLM fine-tuning; multilingual modeling; interpretability; feature attribution; contrastive learning; LLM-based evaluation; safety & robustness analysis
Languages English (native); Mandarin Chinese (HSK Level 5)

SELECTED AWARDS & HONORS

- CSCW 2025 Best Paper Honorable Mention (Top 4%).
- Rising Stars in Data Science (Stanford/UCSD/UChicago); Meta Research PhD Fellow; NSF Graduate Research Fellowship (GRFP); Stanford EDGE Doctoral Fellow; University of Oxford Daniel M. Sachs Class of 1960 Graduating Scholar.
- Charles Ira Young Memorial Tablet & Medal; SEAS Hayes–Palmer Prize; George B. Wood Legacy Prize.